

A COMPACT-RANGE PARTIAL ANSWER TO QUESTION 10.1 OF KALENDA–SPURNY

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ABSTRACT. Kalenda and Spurny ask whether, for a compact convex set X and a Banach space E having the approximation property, or compact approximation property, every affine E -valued Baire-one map on X belongs to the first affine Baire class. We prove a scoped positive result: if E has the approximation property and the range of the affine Baire-one map is norm relatively compact, then the map belongs to $\mathfrak{A}_1(X, E)$.

1. SOURCE QUESTION

The source paper proves the positive result under the bounded approximation property and gives a negative example in a space failing the compact approximation property. It then asks whether the gap can be closed under the weaker approximation-property hypotheses.

In this final section we collect several open questions and discuss which of the results are sharp. We start by pointing out a gap between Theorem 2.2 and Example 2.3.

Question 10.1. *Let X be a compact convex set and E a Banach space having the approximation property (or the compact approximation property). Does any affine function $f \in \mathcal{C}_1(X, E)$ belong to $\mathfrak{A}_1(X, E)$?*

FIGURE 1. Question 10.1 in Kalenda–Spurny [1].

2. RESULT

Theorem 1. *Let X be a compact convex set and let E be a Banach space with the approximation property. Let $f : X \rightarrow E$ be affine and suppose $f \in \mathcal{C}_1(X, E)$. If $f(X)$ is relatively compact in norm, then*

$$f \in \mathfrak{A}_1(X, E).$$

3. IDEA OF THE PROOF

The bounded approximation property in Kalenda–Spurny’s Theorem 2.2 supplies one uniformly bounded sequence of finite-rank operators which converges pointwise on the whole range of f . For a compact range, the ordinary approximation property is already enough: finite-rank operators approximate the identity uniformly on the compact set $\overline{f(X)}$. After composition with f , each approximant has finite-dimensional range, so the source paper’s finite-dimensional Mokobodzki lemma places it in $\mathfrak{A}_1$. The approximants converge back to f inside one bounded ball, and the source paper’s descent result $\mathcal{C}_1 \cap \mathfrak{A}_2 \subset \mathfrak{A}_1$ on bounded convex targets finishes.

4. PROOF

Put $K = \overline{f(X)}^{\|\cdot\|}$. By hypothesis, K is compact. Let

$$M = \sup_{y \in K} \|y\|, \quad R = M + 1.$$

Since E has the approximation property, for each $n \geq 1$ there is a finite-rank operator $L_n : E \rightarrow E$ such that

$$\sup_{y \in K} \|L_n y - y\| < \frac{1}{n}.$$

Define $g_n = L_n \circ f$. Then each g_n is affine, belongs to $\mathcal{C}_1(X, E)$, has finite-dimensional range, and satisfies

$$\|g_n(x)\| \leq R \quad (x \in X).$$

Moreover $g_n \rightarrow f$ uniformly on X .

We next place everything in one bounded convex target. Let B_E denote the closed unit ball of E . Since $f(X) \subset RB_E$ and $f \in \mathcal{C}_1(X, E)$, we also have $f \in \mathcal{C}_1(X, RB_E)$: if $h_m : X \rightarrow E$ are continuous and $h_m \rightarrow f$ pointwise, compose them with the continuous radial retraction

$$\rho(y) = \begin{cases} y, & \|y\| \leq R, \\ Ry/\|y\|, & \|y\| > R \end{cases}$$

from E onto RB_E .

For a fixed n , let $F_n = L_n(E)$, equipped with the norm inherited from E . The map $R^{-1}g_n : X \rightarrow B_{F_n}$ is affine and belongs to $\mathcal{C}_1(X, F_n)$. Kalenda–Spurny’s finite-dimensional lemma [1, Lemma 4.6] therefore gives

$$R^{-1}g_n \in \mathfrak{A}_1(X, B_{F_n}).$$

Multiplying by R , we get $g_n \in \mathfrak{A}_1(X, RB_E)$. Since $g_n \rightarrow f$ pointwise, the definition of the affine Baire hierarchy gives

$$f \in \mathfrak{A}_2(X, RB_E).$$

Finally RB_E is a bounded convex subset of E . The descent corollary of Kalenda–Spurny [1, Corollary 3.8] says that

$$\mathcal{C}_1(X, RB_E) \cap \mathfrak{A}_2(X, RB_E) \subset \mathfrak{A}_1(X, RB_E).$$

Thus $f \in \mathfrak{A}_1(X, RB_E)$, and hence $f \in \mathfrak{A}_1(X, E)$.

5. SCOPE AND OBSTRUCTIONS

The proof uses finite-dimensionality after composing with L_n . This is why the argument does not automatically cover the compact approximation property: compact approximating operators need not have finite-dimensional range. Likewise, without compactness of $f(X)$, the ordinary approximation property does not provide a single sequence of finite-rank operators controlled on the whole range; that is the gap addressed by the bounded approximation property in [1, Theorem 2.2].

6. NOVELTY CHECK

The local run indexes were searched for arXiv:1411.1874, the source title, “Baire classes of affine vector-valued functions”, “approximation property”, “compact approximation property”, “affine Baire-one”, and “Question 10.1”. No existing local packet for this target was found. Bounded web searches for the exact question text and close variants found the source paper and related approximation-property background, but no exact public answer to this compact-range subcase.

REFERENCES

- [1] O. F. K. Kalenda and J. Spurny, *Baire classes of affine vector-valued functions*, arXiv:1411.1874.